



VOICE AI

AI Voice Agents & Telephony Systems

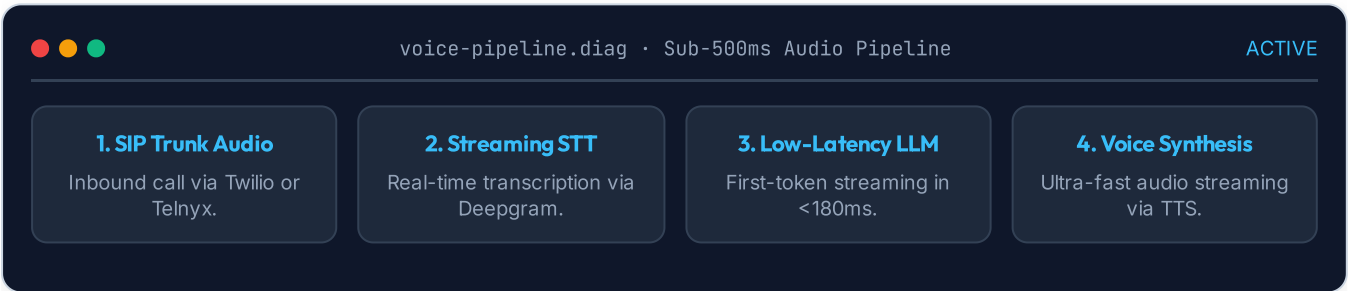
Building sub-500ms real-time conversational voice assistants with WebRTC, SIP trunking, and VAD.

EXECUTIVE SUMMARY

Real-time voice bots require sub-500ms end-to-end turn-taking to feel completely natural. This guide covers streaming speech-to-text, low-latency LLM generation, ultra-fast speech synthesis, and live SIP telephony integration.

1 Streaming Telephony & Audio Pipeline

By streaming audio chunks directly via WebRTC or SIP trunks to a local Voice Activity Detection (VAD) engine, speech processing begins immediately without waiting for long pauses.



2 Production Implementation & Schema

```
// Voice Activity Detection (Silero VAD Event Loop)
audioStream.on('data', chunk => {
  const isSpeaking = vad.process(chunk);
  if (isSpeaking && agentSpeaking) {
    tts.interrupt(); // Immediate barge-in support
    llm.abortCurrentTurn();
  }
});
```



3 Handling Natural Interruptions (Barge-In)

Natural conversation requires allowing callers to interrupt without speech collision. When user speech is detected mid-response, the engine terminates synthesis within 50ms and clears the output queue.

```
call-stream · SIP Session Monitor CONNECTED

[SIP INBOUND] +1 (555) 019-2831 · Connected via Telnyx SIP Trunk
[CALLER] "Hi, I need to schedule an appointment for tomorrow morning."
[STT LATENCY] 140ms · Transcript confirmed
[TOOL EXEC] CheckCalendarAvailability(date: "2026-10-09") → Slot Open
[TTS OUT] "We have a 9:30 AM opening available. Does that time work for you?"
```

4 Key Takeaways & Best Practices

✓ First-Token Priority

Optimize LLM prompts to stream initial conversational acknowledgments ("Sure", "Let me check") within 180ms.

✓ Dedicated SIP Trunks

Use direct SIP trunking rather than browser audio proxies to minimize packet jitter and latency spikes.

✓ Fast Audio Clearing

Ensure the client audio player can drop queued audio frames instantly when an interruption occurs.

✓ Fallback Routing

Always provide a seamless transfer SIP SIP-REFER hook to connect live human agents when required.

IMPLEMENTATION SUPPORT

Need engineering assistance integrating these architectures into your business? Contact our team directly at futureai4all@gmail.com.